



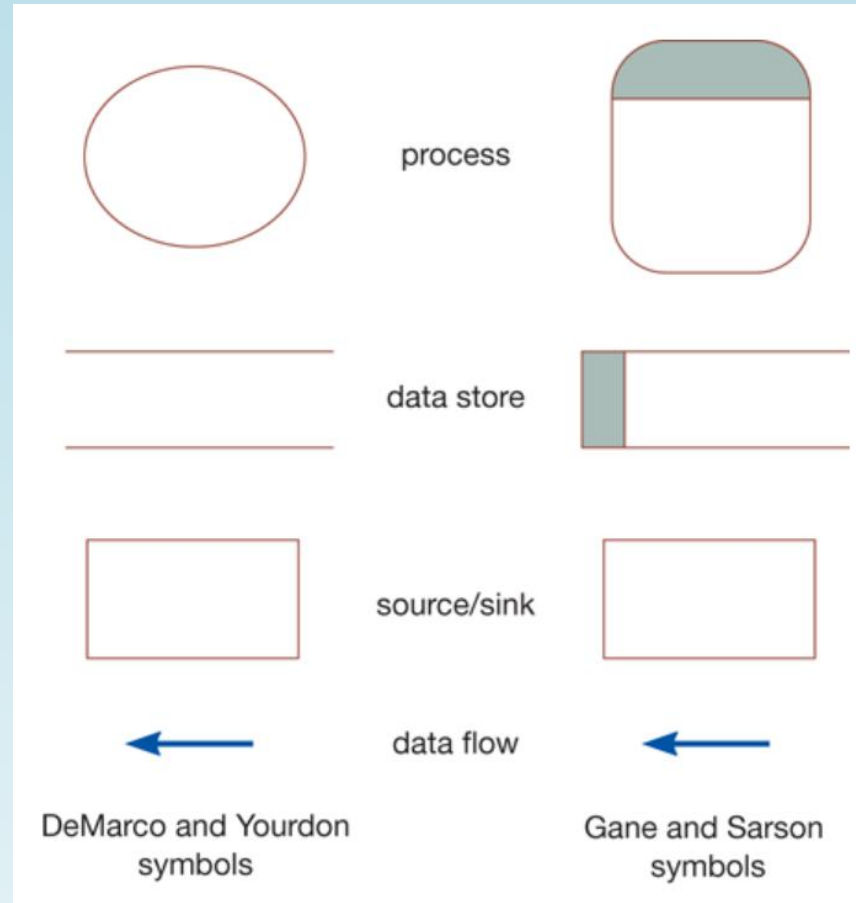
Data Flow Diagram (1)

BIS601

Data Flow Diagram (DFD)

- **DFD**: a picture of the movement of data between external entities and the processes and data stores within a system
- Logical information flows

Definitions and Symbols



Definitions and Symbols

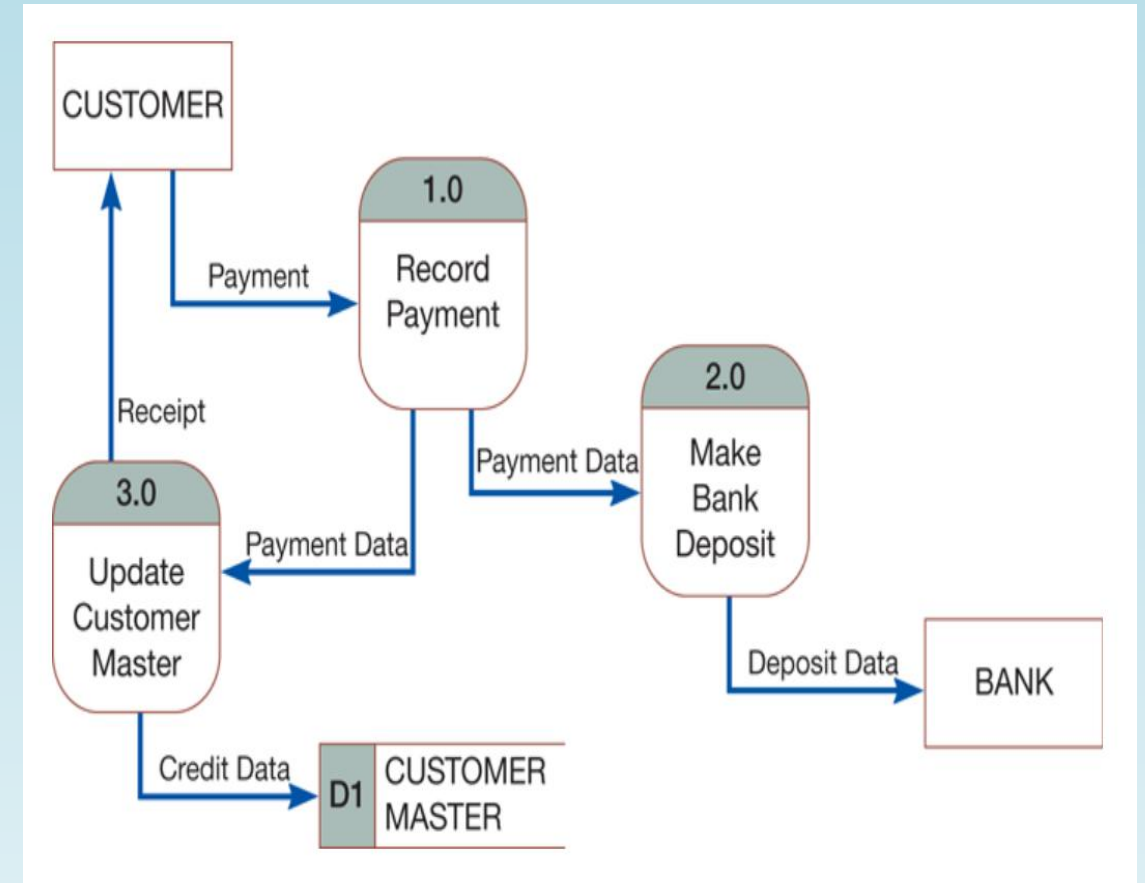
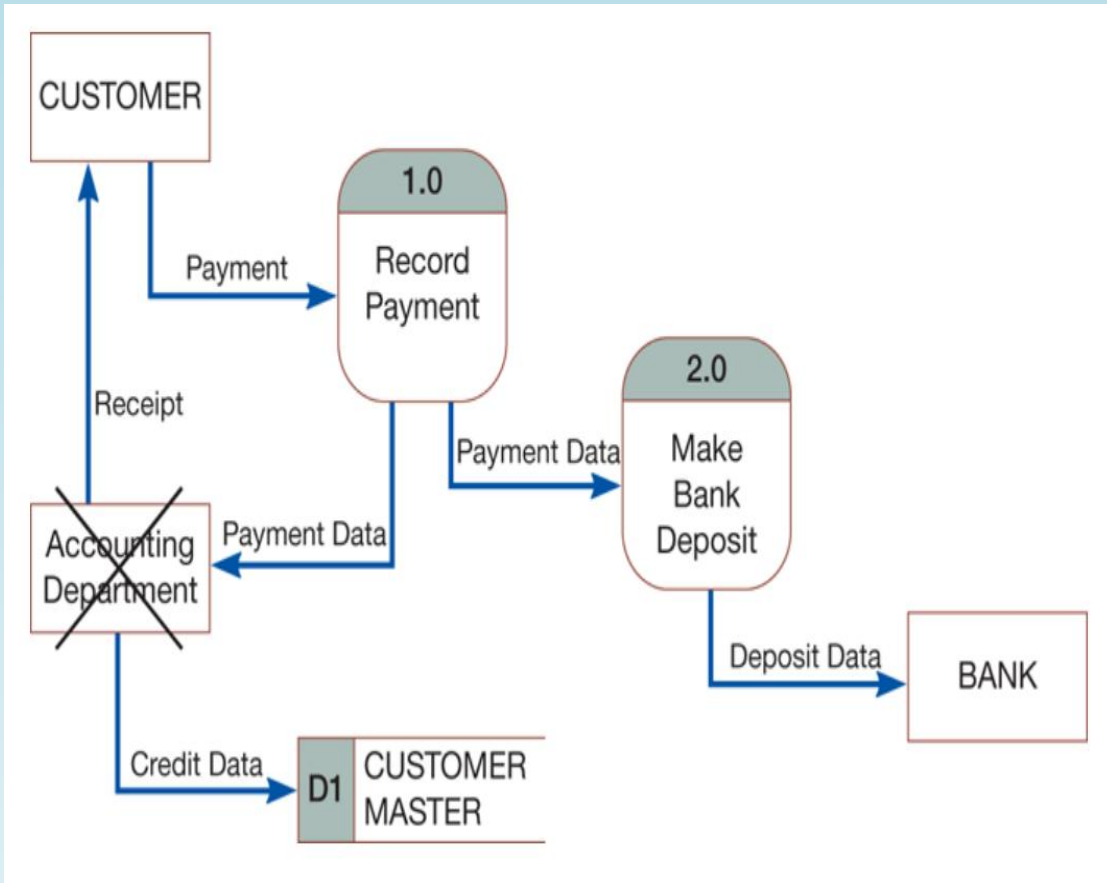
- **Data flow**: data in motion
- **Data store**: data at rest
- **Process**: work or actions performed on data so that they are transformed, stored, or distributed

Definitions and Symbols

- **Source/sink** (external entities): the origin and/or destination of the data
 - Data must originate outside a system from one or more sources, and the system must produce information to one or more sinks

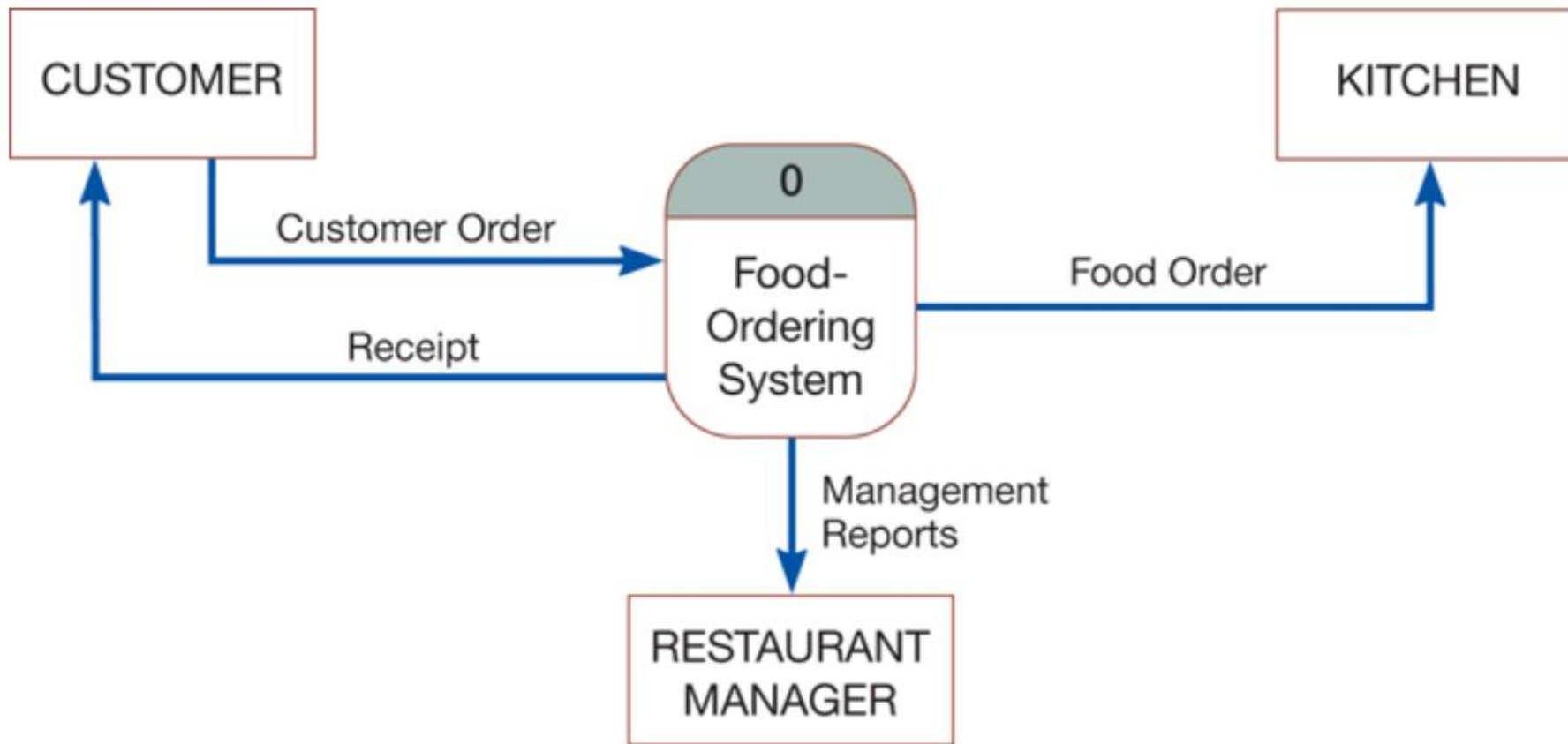
Definitions and Symbols

- A source/sink
 - Organization or organization unit e.g., a supplier, HR department
 - Person (role) who interacts with the system e.g., a customer
 - Another information system with which the system you are analyzing exchanges information



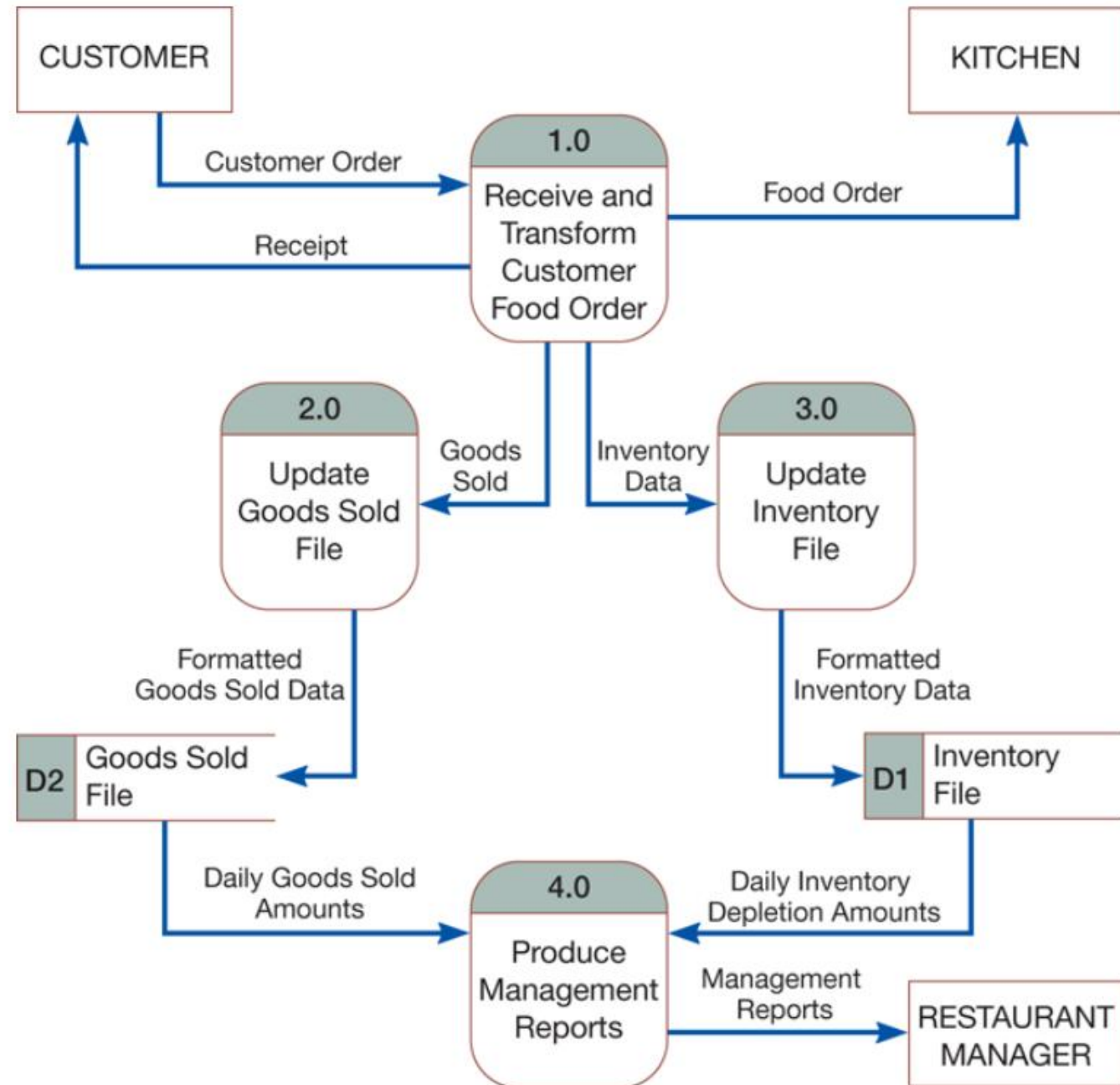
Context Diagram

- **Context diagram**: an overview of a system that shows the system boundaries, external entities that interact with the system, and the major information flows between the entities and the system



Level-0 Diagram

- **Level-0 diagram:** a DFD that represents a system's major processes, data flows, and data stores at a high level of detail



Rules

Process:

- A. No process can have only outputs. It would be making data from nothing (a miracle). If an object has only outputs, then it must be a source.
- B. No process can have only inputs (a black hole). If an object has only inputs, then it must be a sink.
- C. A process has a verb phrase label.

Data Store:

- D. Data cannot move directly from one data store to another data store. Data must be moved by a process.
- E. Data cannot move directly from an outside source to a data store. Data must be moved by a process that receives data from the source and places the data into the data store.
- F. Data cannot move directly to an outside sink from a data store. Data must be moved by a process.
- G. A data store has a noun phrase label.

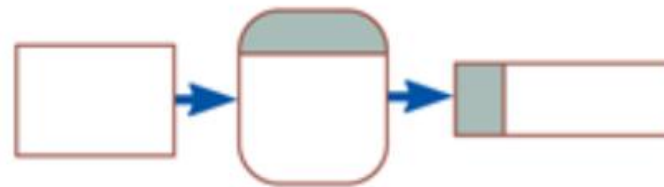
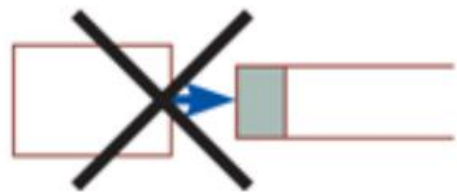
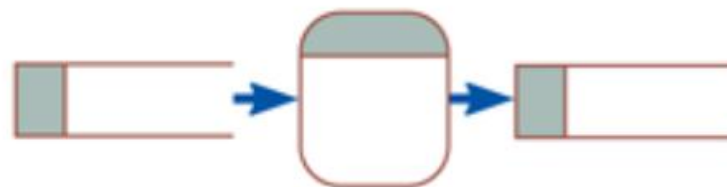
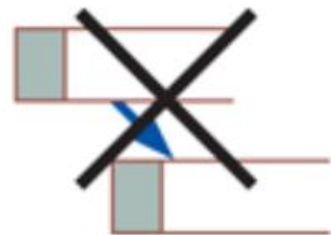
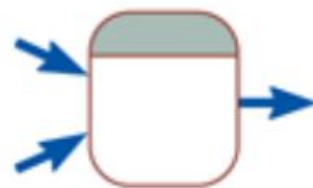
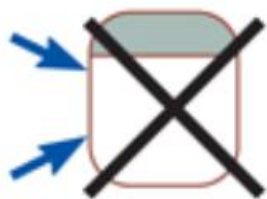
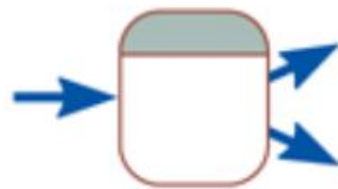
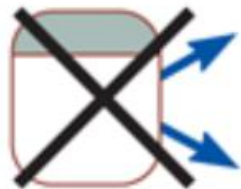
Source/Sink:

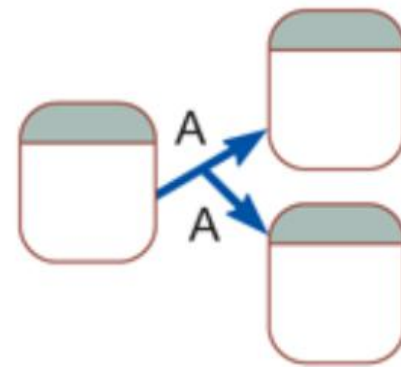
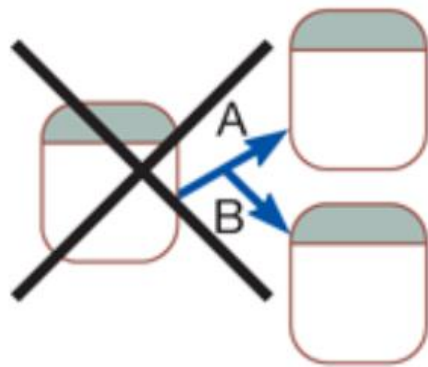
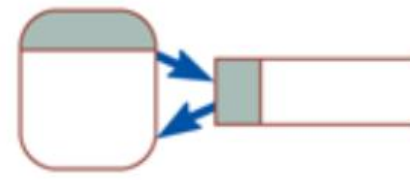
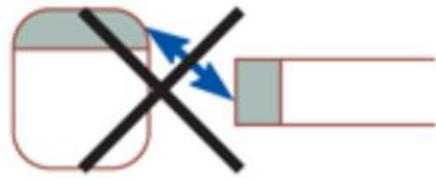
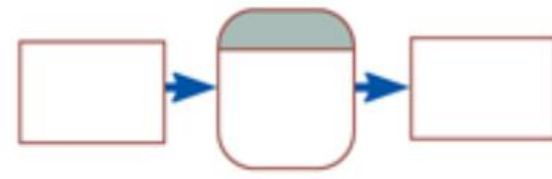
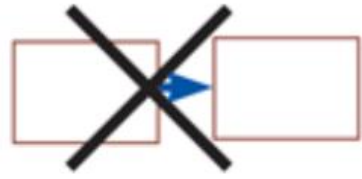
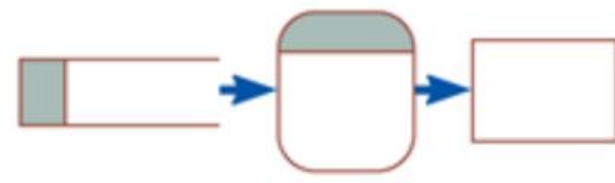
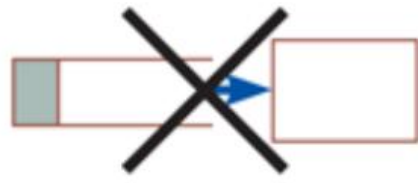
- H. Data cannot move directly from a source to a sink. It must be moved by a process if the data are of any concern to our system. Otherwise, the data flow is not shown on the DFD.
- I. A source/sink has a noun phrase label.

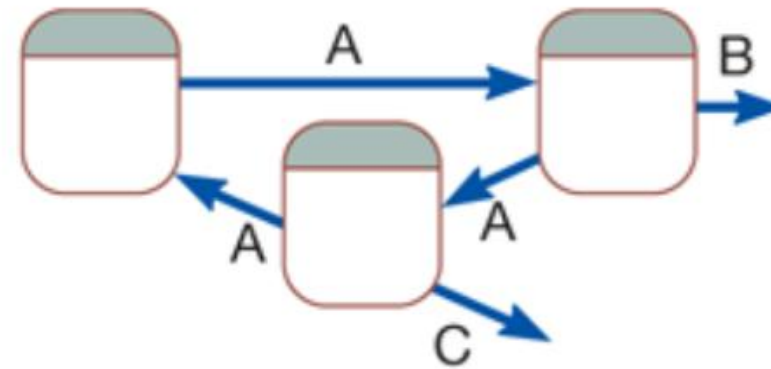
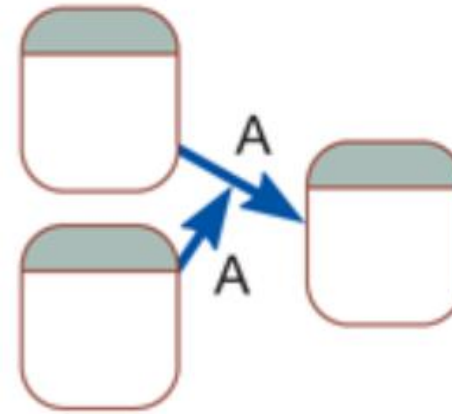
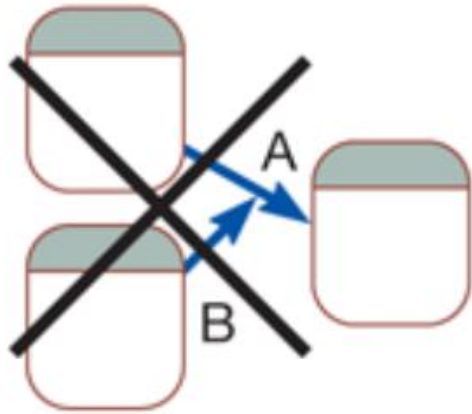
Rules

Data Flow:

- J. A data flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The latter is usually indicated, however, by two separate arrows because these happen at different times.
- K. A fork in a data flow means that exactly the same data goes from a common location to two or more different processes, data stores, or sources/sinks (this usually indicates different copies of the same data going to different locations).
- L. A join in a data flow means that exactly the same data come from any of two or more different processes, data stores, or sources/sinks to a common location.
- M. A data flow cannot go directly back to the same process it leaves. There must be at least one other process that handles the data flow, produces some other data flow, and returns the original data flow to the beginning process.
- N. A data flow to a data store means update (delete or change).
- O. A data flow from a data store means retrieve or use.
- P. A data flow has a noun phrase label. More than one data flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.





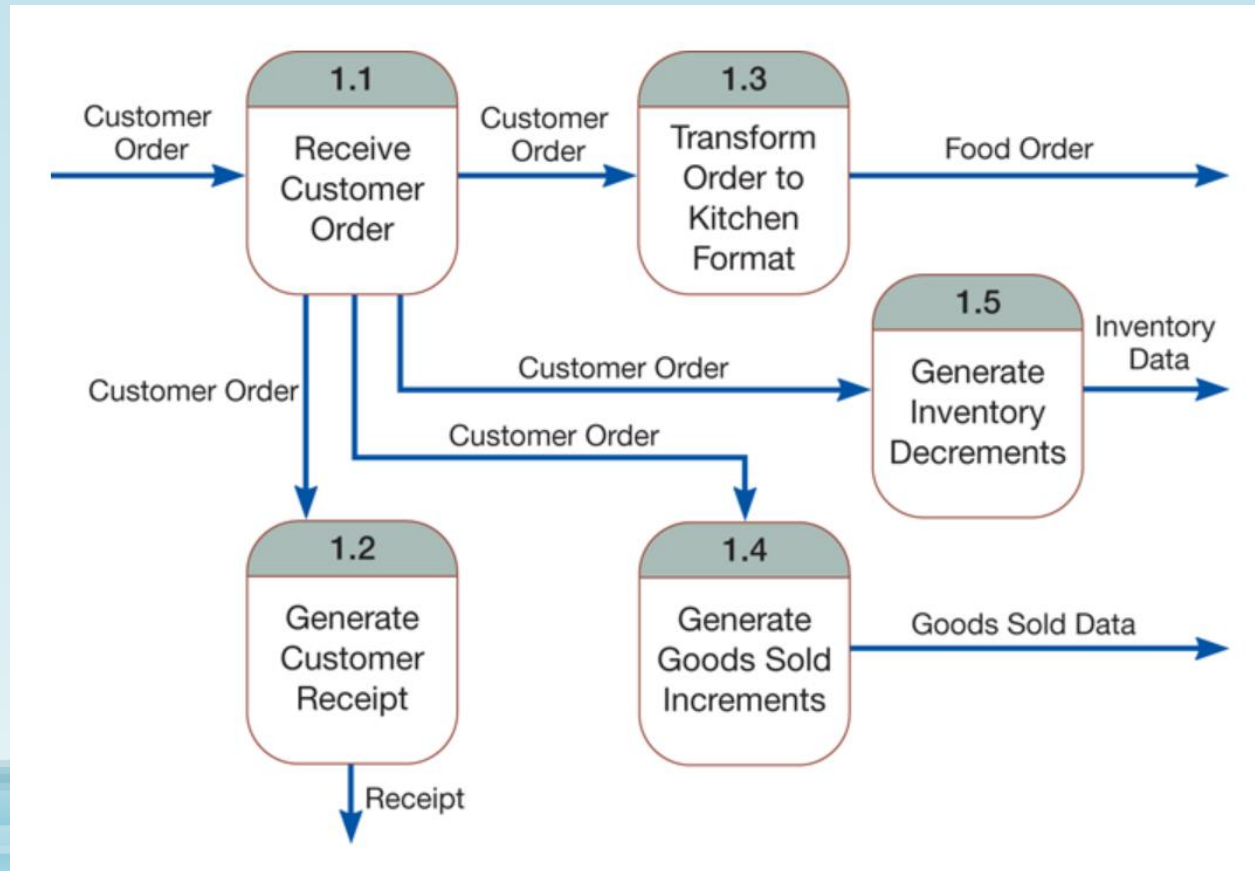


Decomposition of DFDs

- **Functional decomposition:** an iterative process of breaking the description of a system down into finer and finer detail, which creates a set of charts in which one process on a given chart is explained in greater detail on another chart

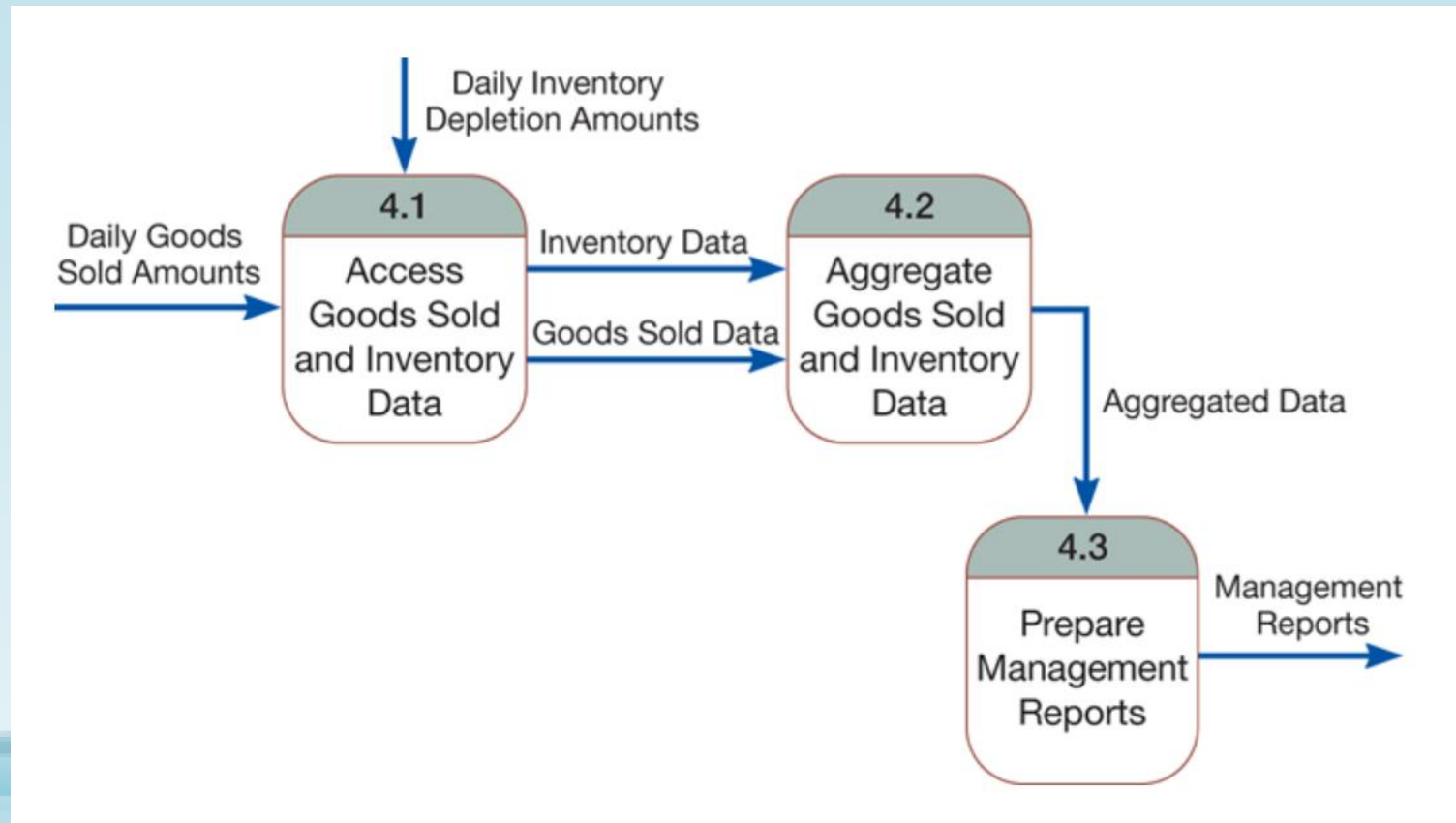
Level-1 Diagram

- Level-1 diagram of process 1



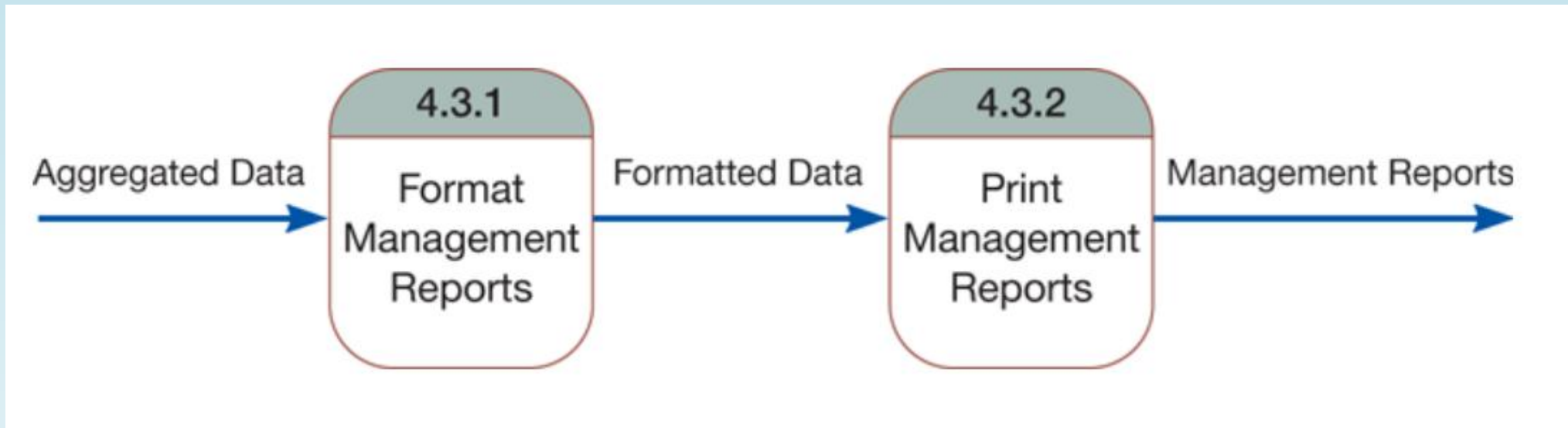
Level-1 Diagram

- Level-1 diagram of process 4



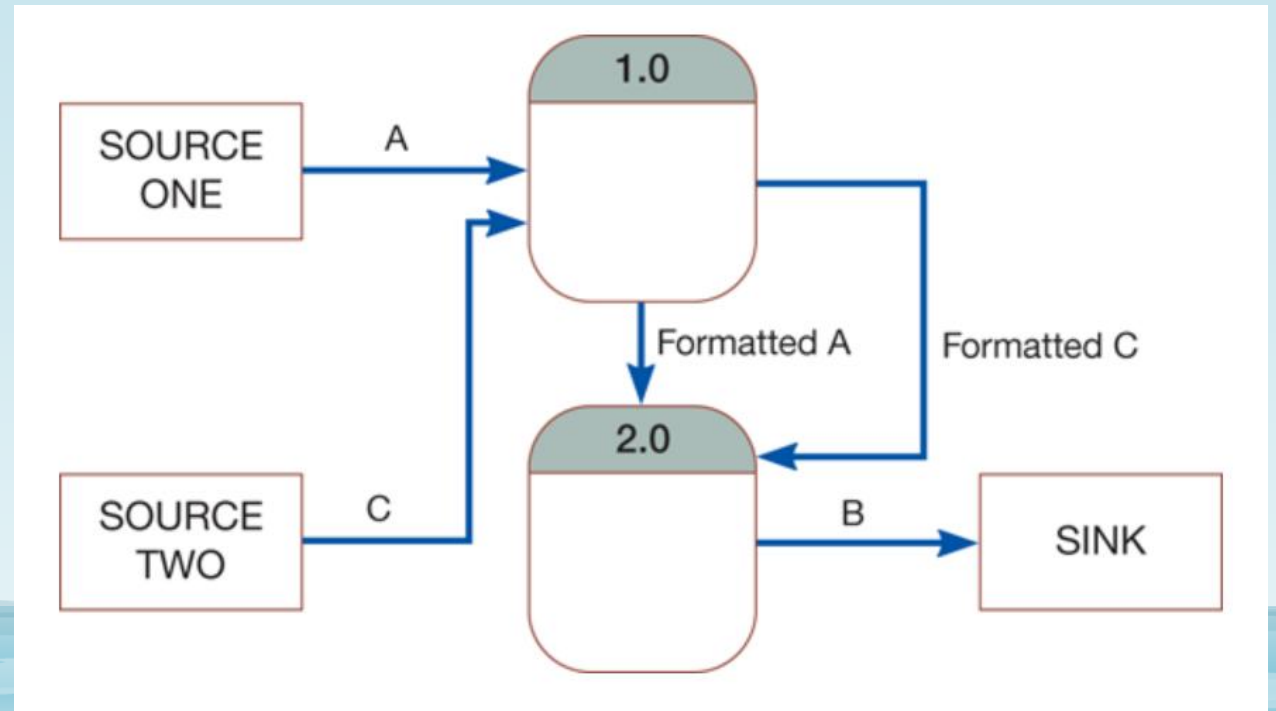
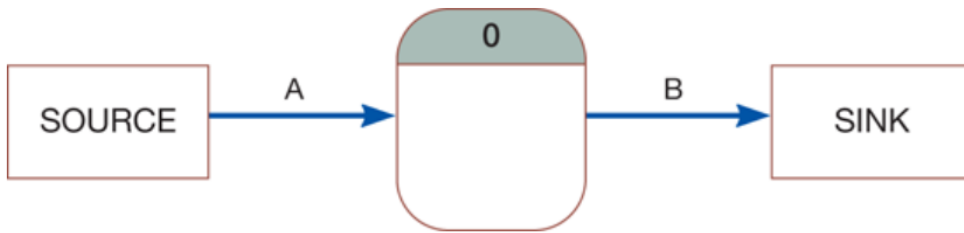
Level-2 Diagram

- Level-2 diagram of process 4.3



Balancing DFDs

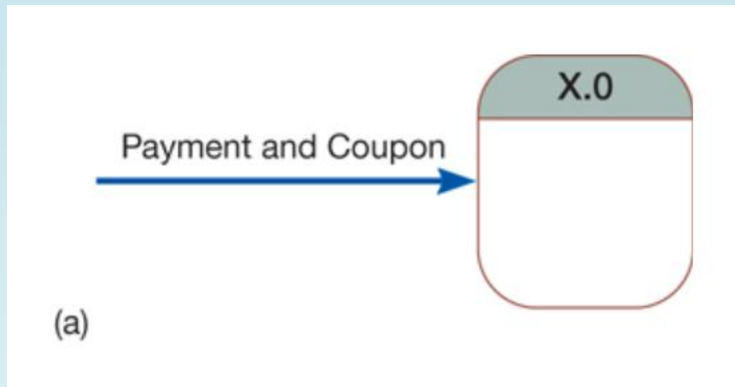
- Unbalanced



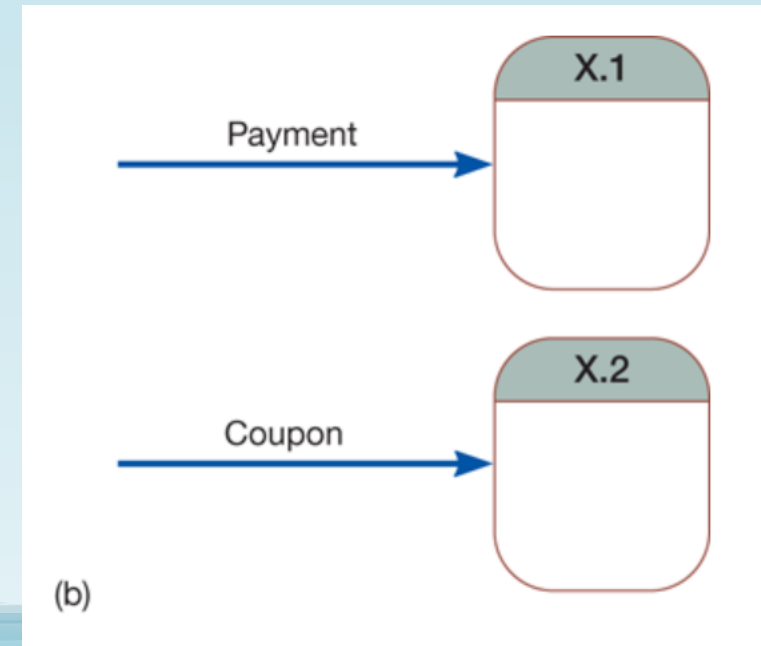
Balancing DFDs

- Balanced

(a) Composite data flow



(b) Disaggregated data flows



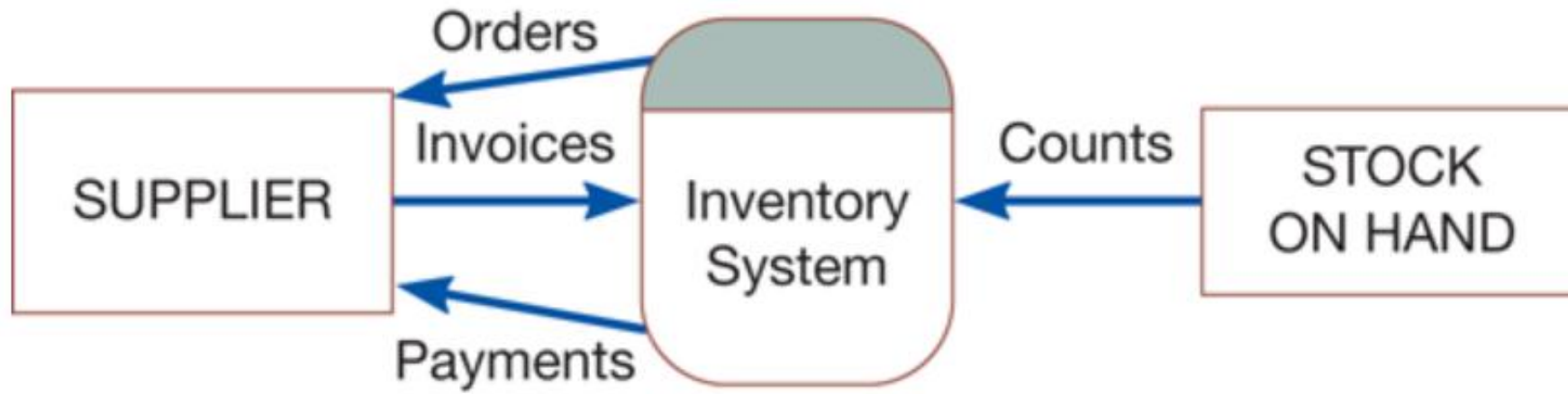
Example

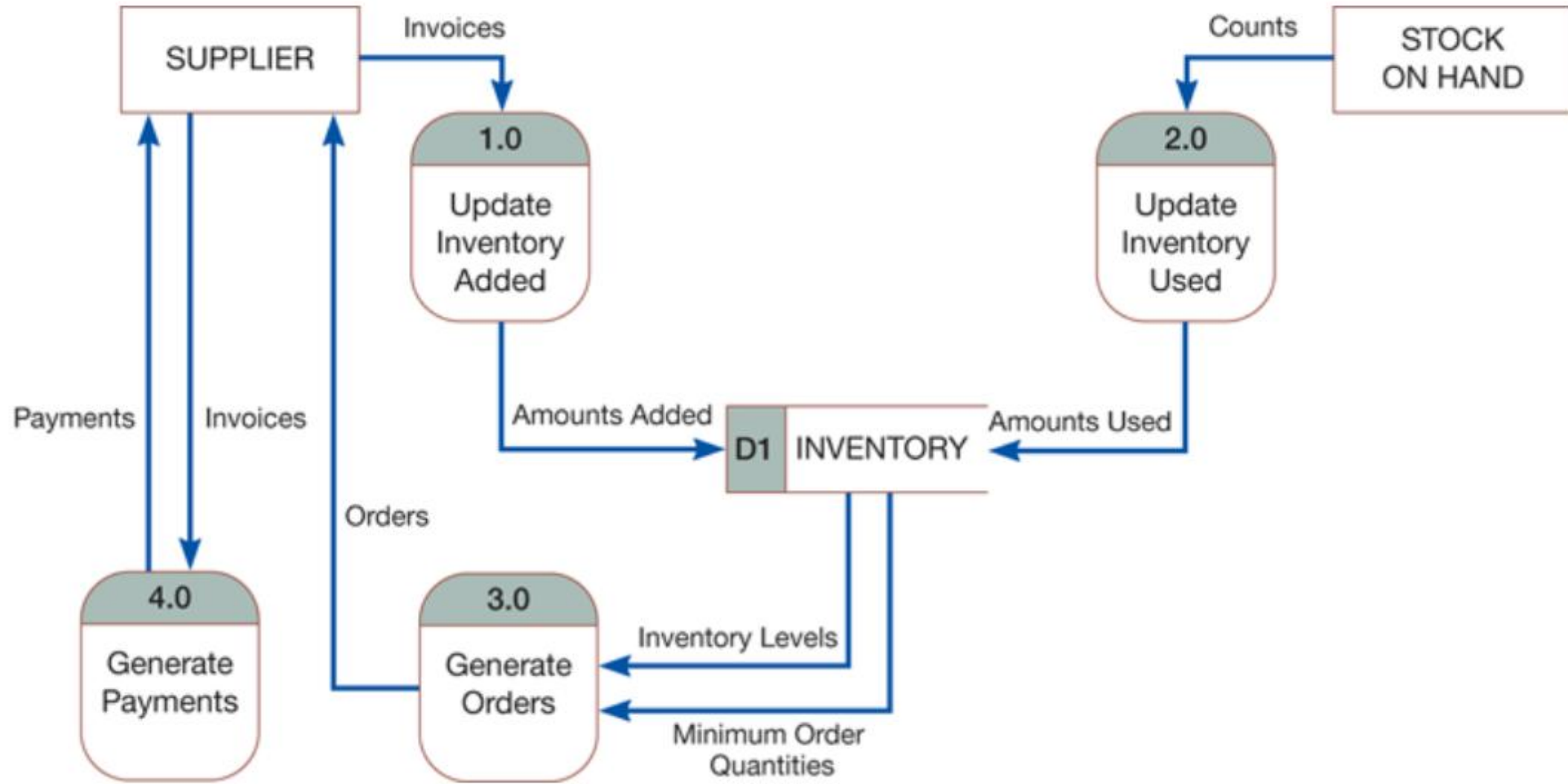
- Inventory system

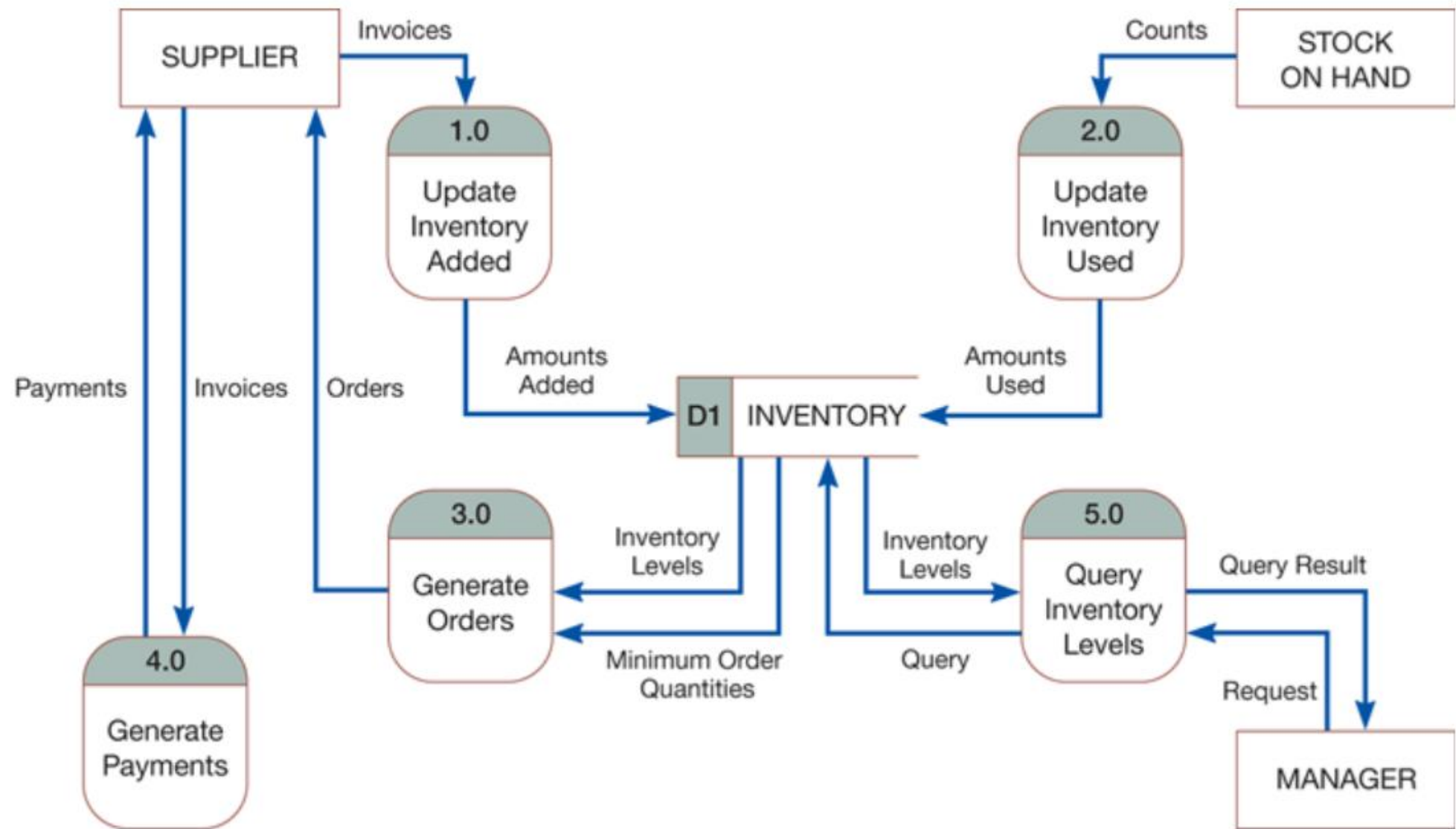
1. Meet delivery trucks before opening restaurant.
2. Unload and store deliveries.
3. Log invoices and file in accordion file.
4. Manually add amounts received to stock logs.
5. After closing, print inventory report.
6. Count physical inventory amounts.
7. Compare inventory report totals to physical count totals.
8. Compare physical count totals to minimum order quantities. If the amount is less, make order; if not, do nothing.
9. Pay bills that are due and record them as paid.

Example

- Four key processes
 - Account for anything added to inventory
 - Account for anything taken from inventory
 - Place orders
 - Pay bills







Guidelines for Drawing DFDs

- **Completeness:** all necessary components of a DFD have been included and fully described
- **Consistency:** balancing
- **Timing:** no indication
- **Iterative DFDs development**

Guidelines for Drawing DFDs

- **Primitive DFDs:** the lowest level of decomposition for a DFD
 - A single decision, calculation, or database operation